

# Guide: Environment Setup for Huawei Cloud Development



Huawei Technologies Co., Ltd.

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# 1 Provision a Flexus X Instance on Huawei Cloud

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**General Objective:** Provision a Flexus X instance on Huawei Cloud that will host a complete coding stack environment: the [oh-my-coding-maas-gateway](#) (a LiteLLM-based proxy with load balancing, virtual keys, and Grafana observability), governance tooling, and multiple coding agents — with Open-Code as the primary one. By the end of this chapter you will have a running Flexus X instance with a public IP accessible over SSH.

## Prerequisites:

- Active Huawei Cloud account with IAM credentials.
- Permission to create security groups, Flexus X instances, and EIPs.
- A key pair created in the Console, or permission to create one.
- A web browser (Chrome, Firefox, or Edge).

### Info

This guide assumes the `sa-brazil-1` region. Adjust the region if your account is registered elsewhere.

## 1.1 Create the security group (sg-xgate)

**Objective:** Create a security group that allows SSH access on port 4444 from specific Huawei internal IPs only.

### Info

We use port 4444 instead of the standard SSH port 22 because the Huawei corporate proxy blocks outbound connections on port 22. Port 4444 is allowed through the proxy and works reliably for SSH.

## Step by step:

1. In the Console, go to menu:VPC[Security Groups] and click **Create Security Group**.
2. Name it sg-xgate and add the following inbound rules:

```
Rule 1:
  Protocol:    TCP
  Port:        4444
  Source:      119.8.89.193/32
  Description: SSH for Huawei internal user 1
```



```
Rule 2:
  Protocol:    TCP
  Port:        4444
  Source:      119.8.89.3/32
  Description: SSH for Huawei internal user 2

Rule 3:
  Protocol:    TCP
  Port:        4444
  Source:      119.8.193.3/32
  Description: SSH for Huawei internal user 3
```

3. Click **OK** to create the security group.

#### Tip

Each /32 rule restricts access to exactly one IP address. Only the three listed Huawei internal IPs can reach the instance on port 4444 — no other inbound traffic is allowed.

## 1.2 Create the Flexus X instance

**Objective:**– Launch a single Flexus X instance using the Console wizard, using the security group created above.

#### Step by step:

1. Log in to the [Huawei Cloud Console](#) and select the `sa-brazil-1` region in the top-right corner.
2. Open the service catalog and select **Flexus X** (under **Compute**).
3. Click **Create Flexus X** to open the creation wizard.
4. Under **Billing Mode**, select **Pay-per-use**.

#### Tip

Pay-per-use bills by the hour with no long-term commitment. You only pay for the time the instance is running. This is ideal for development and testing — stop the instance when not in use to reduce costs.

1. Set the compute resources using the slider: drag to **4 vCPU** and **16 GB RAM**.

#### Info

Flexus X uses a slider for vCPU and memory instead of fixed flavor names. The resulting flavor ID is `x1.4u.16g`, but you select it by dragging the vCPU and RAM controls — there is no flavor dropdown.

1. Fill in the remaining basic configuration:

AZ:                      Any (choose the availability zone you prefer)



|                 |                                                |
|-----------------|------------------------------------------------|
| Image:          | Ubuntu 24.04 Server 64bit                      |
| Disk:           | 100 GB General Purpose SSD                     |
| VPC:            | vpc-default                                    |
| Subnet:         | subnet-default                                 |
| Security Group: | sg-xgate (port 4444, Huawei internal IPs only) |

### Important

The EIP is auto-assigned and configured to be released when the instance is deleted. If you need a fixed IP that persists across stop/start cycles, allocate a dedicated EIP and bind it manually after creation.

1. Under **EIP** settings, configure the public IP:

|            |                                            |
|------------|--------------------------------------------|
| EIP:       | Auto-assign                                |
| Billing:   | Pay by traffic                             |
| Bandwidth: | 1000 Mbit/s                                |
| Release:   | Yes – release EIP when instance is deleted |

2. Under **Cloud Eye**, enable monitoring (recommended).

### Info

Cloud Eye is a monitoring service that provides functions such as real-time monitoring, alarm reporting, resource grouping, and website monitoring. Enabling it gives you visibility into instance health, CPU, memory, and network metrics out of the box.

1. Set the instance name to `flexusx-dev` (or any name you prefer).
2. Under **Login Configuration**, select **Key pair** as the authentication method. If you do not yet have a key pair, click **Create Key Pair** to generate one inside Huawei Cloud and download the `.pem` file.

### Info

The key pair is the primary access method. Later, you can set a root password on the instance (via the Console or after first login with the key) to have a second way to access the instance.

1. Click **Create Now** and wait for the status to change to **Running**.

## 1.3 Verify the instance

On the instance details page, copy the **EIP** field. This is the address you will use to connect in the next chapters.



| Step           | What to check                | Expected result      |
|----------------|------------------------------|----------------------|
| Security group | sg-xgate with TCP 4444 rules | 3 inbound rules      |
| Billing mode   | Pay-per-use selected         | Hourly billing       |
| Compute        | 4 vCPU, 16 GB RAM via slider | Flavor x1 . 4u . 16g |
| Image          | Ubuntu 24.04 Server 64bit    | Selected             |
| EIP            | Auto-assigned                | Public IP available  |
| Cloud Eye      | Monitoring enabled           | Metrics visible      |
| Instance name  | fLexusx-dev                  | Set                  |
| Key pair       | Created and downloaded       | . pem file saved     |
| Instance       | Status is <b>Running</b>     | EIP copied           |

**Table 1:** Chapter 1 verification checklist.

# 2

## Configure Flexus X for Remote Access

---

**General Objective:** Prepare the Flexus X instance for remote access by setting a root password via the Console and changing the SSH port from 22 to 4444. This is required because the Huawei corporate proxy blocks outbound connections on port 22.

**Prerequisites:**

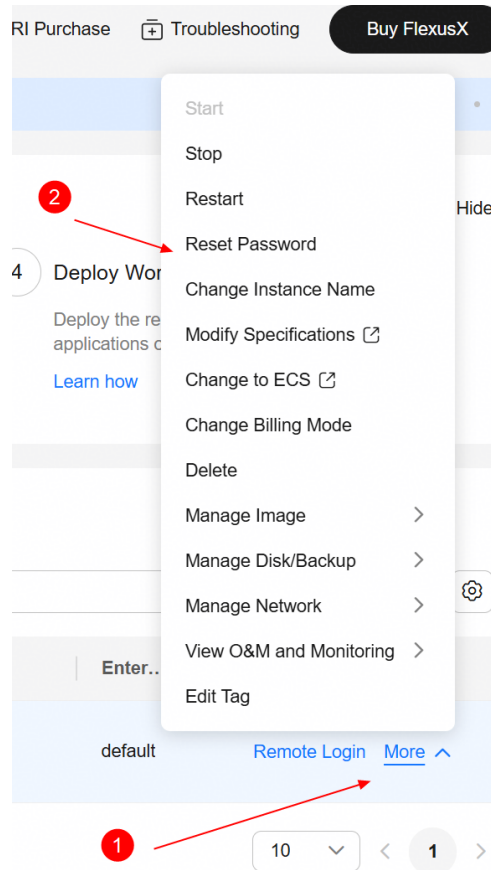
- Chapter 1 completed — Flexus X instance running with a public IP.
  - Access to the Huawei Cloud Console.
- 

### 2.1 Set the root password

**Objective:** Use the Console to reset the root password of the Flexus X instance.

**Step by step:**

1. In the Console, navigate to **Flexus X** and click on the instance `f1exusx-dev`.
2. In the instance details page, click menu:More[Reset Password].



**Figure 1:** Reset Password option in the Flexus X details page.

3. Enter a strong root password and confirm. Click **OK**.
4. The instance will restart automatically. Wait for the status to return to **Running**.

## Important

Choose a strong password (12+ characters, mixed case, digits, symbols). This password enables remote login directly from the Console and serves as a second access method alongside the SSH key pair.

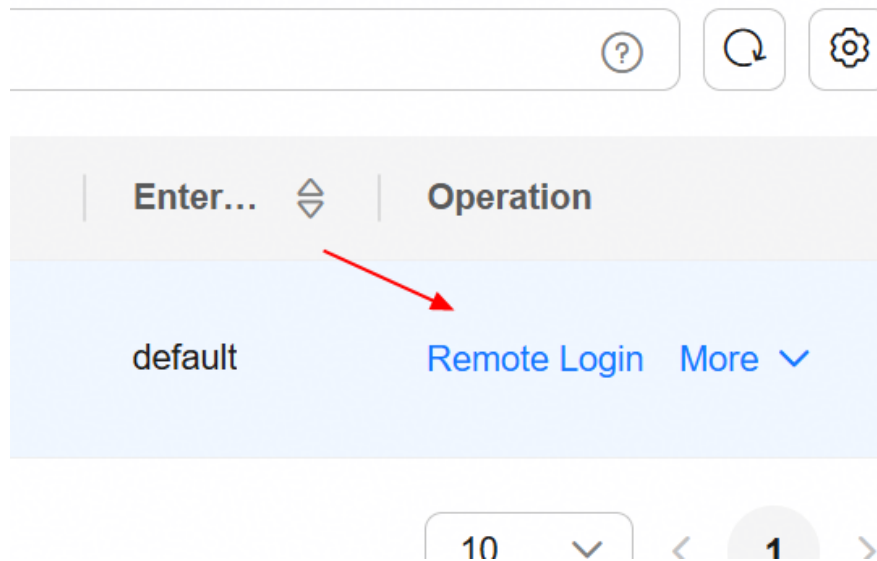
## 2.2 Change the SSH port to 4444

**Objective:** Log in via the Console remote login, edit `sshd\_config`, and change the SSH port from 22 to 4444.

### Step by step:

1. In the instance details page, click **Remote Login**. A browser-based terminal opens.





**Figure 2:** Remote Login button in the Flexus X details page.

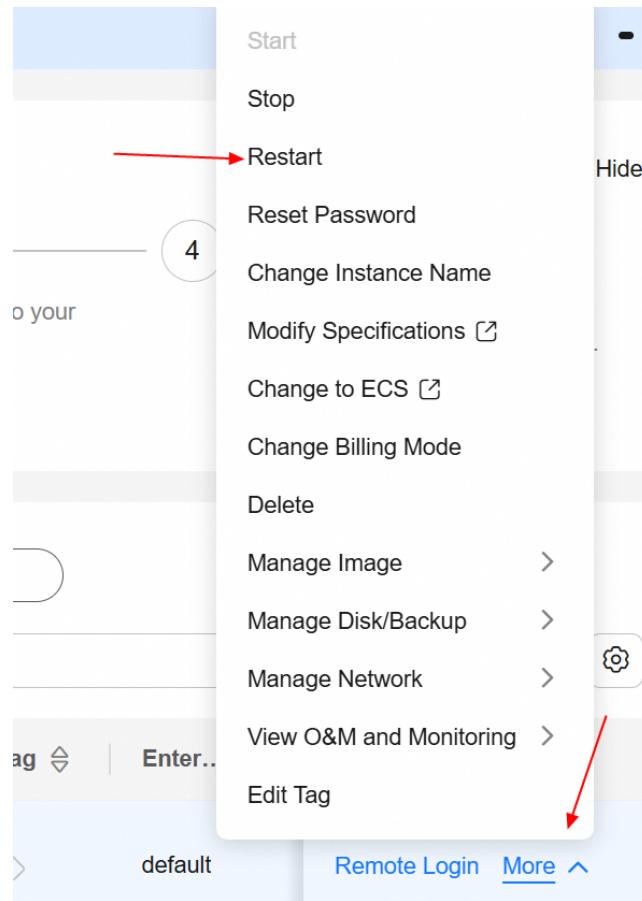
2. Log in as root using the password set in the previous subsection.
3. Open the SSH daemon configuration file:

```
sudo vim /etc/ssh/sshd_config
```

4. Find the line Port 22 (or #Port 22) and change it to:

```
Port 4444
```

5. Save and exit vim (:wq).
6. Reboot the instance from the Console to apply the change. The SSH daemon will start automatically on port 4444 after reboot.



**Figure 3:** Restart the Flexus X instance from the Console.

+ WARNING: The reboot is **mandatory** — not optional. Until the instance is rebooted, SSH continues listening on port 22 and the proxy CONNECT to port 4444 will fail with Ncat: Error reading proxy response Status-Line. If you skip this step, all subsequent SSH connections through the proxy will fail.

#### Info

After this change, SSH connections must use Port 4444 explicitly. The security group sg-xgate (created in Chapter 1) already allows inbound TCP 4444 from the authorized Huawei internal IPs.

## 2.3 Verify the configuration

Use the table below to confirm all steps were completed successfully before proceeding to the next chapter:



| Step             | What to check                                    | Expected result |
|------------------|--------------------------------------------------|-----------------|
| Security group   | <code>sg-xgate</code> exists with TCP 4444 rules | 3 inbound rules |
| Flexus X running | Instance <code>flexusx-dev</code> status         | <b>Running</b>  |
| Root password    | Remote Login works with password                 | Shell prompt    |
| SSH port         | <code>sshd\_config</code> has Port 4444          | Config saved    |
| SSH restart      | Instance rebooted from Console                   | Service active  |

**Table 2:** Chapter 2 verification checklist.

# 3

## Install ncat and SSH Connection Topology

**General Objective:** Install ncat (from Nmap) on your local Windows machine and understand the SSH connection topology through the Huawei corporate proxy (xgate). ncat serves as the SSH ProxyCommand, tunneling SSH through the HTTP proxy via the CONNECT method.

**Prerequisites:**

- Chapter 2 completed — Flexus X configured with root password and SSH on port 4444.
- The private key file (.pem) downloaded from Huawei Cloud.

### Info

**Premises:**

- The proxy is **xgate** (Huawei corporate proxy) at `proxy.huawei.com:8080`.
- SSH runs on port `4444` (configured in Chapter 2).
- Login as root using the key pair from Chapter 1.
- ncat (from Nmap) is used as the ProxyCommand to tunnel SSH through the HTTP proxy via the CONNECT method.
- Proxy authentication is transparent — it is handled by the corporate VPN, not by credentials in the SSH config.

### 3.1 How the connection works

When you specify a ProxyCommand in the SSH config, SSH launches ncat as a child process. ncat performs the HTTP CONNECT handshake with the proxy:

```
ncat -> proxy:    CONNECT <EIP>:4444 HTTP/1.1
                  Host: <EIP>:4444

proxy -> ncat:    HTTP/1.1 200 Connection Established

[Both sides now send raw TCP bytes -- SSH negotiation begins over the tunnel]
```

After the `200 Connection Established` response, the proxy becomes a blind TCP relay — it neither inspects nor modifies the SSH traffic. SSH key exchange, encryption, and authentication all happen inside this tunnel as if it were a direct TCP connection.

**Info**

The corporate proxy blocks `CONNECT` to port 22 (standard SSH) but allows it to port 4444. Enterprise HTTP proxies implement a `CONNECT` port allowlist: port 443 (HTTPS) is always allowed, port 22 is usually blocked, and port 4444 is often allowed because it is commonly used for HTTPS-alt. This is why we configured SSH on port 4444 in Chapter 2.

## 3.2 Why ncat

ncat is the chosen proxy tunnel tool for several reasons:

- **Full HTTP CONNECT support** — ncat implements the `CONNECT` method natively via `--proxy-type http`.
- **Actively maintained** — part of the Nmap project, with regular releases and bug fixes.
- **IPv4 forcing** — the `-4` flag prevents failed IPv6 attempts through the proxy.
- **Connect timeout** — the `-w` flag prevents indefinite hangs if the proxy is unreachable.
- **Proxy auth support** — `--proxy-auth` and the `NCAT\_PROXY\_AUTH` environment variable, if explicit credentials are ever needed.

**Info**

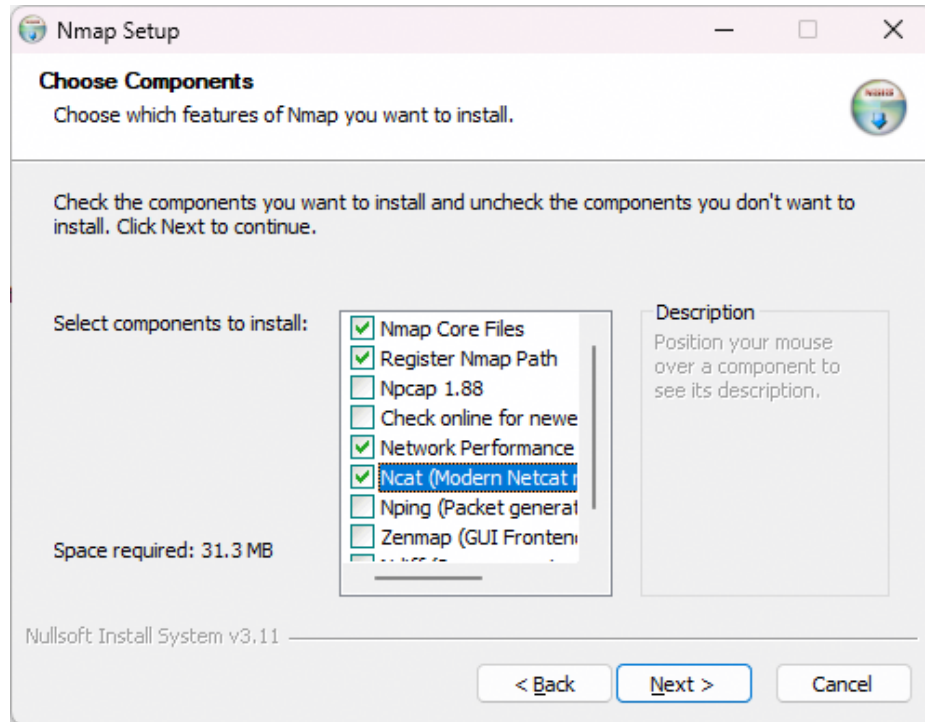
`connect.exe` (from Git for Windows) is a lighter alternative that ships with Git, but it lacks the `-4` (force IPv4) and `-w` (connect timeout) flags, which are important for reliable proxied connections. ncat provides better control over the tunnel behavior.

## 3.3 Install ncat on Windows

**Objective:** Install Nmap, which provides `ncat.exe` — the proxy tunnel tool used by SSH ProxyCommand.

**Step by step:**

1. Download the Nmap installer from [nmap.org/download.html](https://nmap.org/download.html). Use the **Latest stable release self-installer** (`nmap-7.991-setup.exe`, tested and verified).
2. Run the installer (right-click **Run as Administrator**).
3. During installation, uncheck **Npcap** — it is a kernel driver for packet capture and is not needed for proxy tunneling. Also uncheck **Zenmap**, **Nping**, and **Ndiff** for a minimal install.



**Figure 4:** Nmap installer component selection — uncheck Npcap, Zenmap, Nping, and Ndiff.

+ TIP: Skipping Npcap avoids a kernel driver install and potential reboot. ncat in proxy mode uses standard Winsock2 TCP sockets and does not need Npcap at all.

4. Choose the default install path: C:\Program Files (x86)\Nmap.
5. After installation, verify that ncat.exe is available:

```
"C:\Program Files (x86)\Nmap\ncat.exe" --version
```

6. If the command shows the version string, the installation is correct.

### Important

Nmap is commonly flagged by antivirus software due to port-scanning heuristics. Windows Defender may quarantine ncat.exe on download or block execution. Add an exclusion for the Nmap folder and the ncat.exe process in menu: Windows Security[Virus & threat protection, Exclusions]. If ncat.exe was already quarantined, restore it from Protection history before adding the exclusion.

| Step          | What to check                         | Expected result      |
|---------------|---------------------------------------|----------------------|
| Nmap download | nmap-7.991-setup.exe saved            | Installer ready      |
| Components    | Npcap, Zenmap, Nping, Ndiff unchecked | Minimal install      |
| Install path  | C:\Program Files (x86)\Nmap           | Default path         |
| ncat.exe      | ncat --version shows version          | Installed correctly  |
| Antivirus     | Exclusion added for Nmap folder       | ncat.exe not blocked |

**Table 3:** Chapter 3 verification checklist.

# 4 Install VS Code and Connect via Remote-SSH

---

**General Objective:** Install VS Code from scratch on your local Windows machine, install the Remote-SSH extension, configure the SSH config file using VS Code as the editor, and connect to the Flexus X instance.

**Prerequisites:**

- Chapter 3 completed — ncat installed and proxy tunnel understood.
  - The private key file (.pem) downloaded from Huawei Cloud.
- 

## 4.1 Install VS Code

**Step by step:**

1. Download VS Code from [code.visualstudio.com/download](https://code.visualstudio.com/download). Choose the **Windows** installer (User Installer, 64-bit).
2. Run the installer with the default settings.
3. After installation, verify VS Code is available:

```
code --version
```

## 4.2 Install the Remote-SSH extension

**Step by step:**

1. Open VS Code on your local Windows machine.
2. Install the **Remote - SSH** extension (publisher: Microsoft):

```
code --install-extension ms-vscode-remote.remote-ssh
```

## 4.3 Configure the SSH config file

**Objective:** Create an SSH config entry that tunnels through the Huawei proxy using ncat from Nmap. Use VS Code as the editor.

### Step by step:

1. Open or create the SSH config file using VS Code:

```
code %USERPROFILE%\.ssh\config
```

2. Add the following configuration. Replace these placeholders with your values:

- `<host-name>` — a name of your choice (e.g., `flexusx-dev`)
- `<EIP>` — the Flexus X public IP copied from Chapter 1
- `<user>` — your Windows username (e.g., `w00946241`)
- `<key-file>` — the name of your `.pem` key file downloaded from Huawei Cloud (e.g., `KeyPair-opencode`)

```
Host <host-name>
  HostName <EIP>
  User root
  Port 4444
  IdentityFile "C:\Users\<user>\.ssh\<key-file>.pem"
  ProxyCommand "C:\Program Files (x86)\Nmap\ncat.exe" -4 -w 10 --proxy
    proxy.huawei.com:8080 --proxy-type http %h %p
  ServerAliveInterval 15
  ServerAliveCountMax 4
  ConnectTimeout 10
  TCPKeepAlive yes
  StrictHostKeyChecking accept-new
```

#### Info

The `IdentityFile` path has two placeholders: `<user>` is your Windows username (the folder under `C:\Users\`), and `<key-file>` is the name of the `.pem` file downloaded from Huawei Cloud in Chapter 1. For example: `C:\Users\w00946241\.ssh\KeyPair-opencode.pem`.



## 4.4 What each ncat flag does

| Flag              | Purpose                                                       |
|-------------------|---------------------------------------------------------------|
| -4                | Force IPv4 (avoids failed IPv6 attempts through the proxy)    |
| -w 10             | 10-second connect timeout (prevents indefinite hang)          |
| --proxy           | Proxy address and port ( <code>proxy.huawei.com:8080</code> ) |
| --proxy-type http | Use HTTP CONNECT method                                       |
| %h                | Substituted by SSH with the HostName (the EIP)                |
| %p                | Substituted by SSH with the Port (4444)                       |

**Table 4:** ncat ProxyCommand flags.

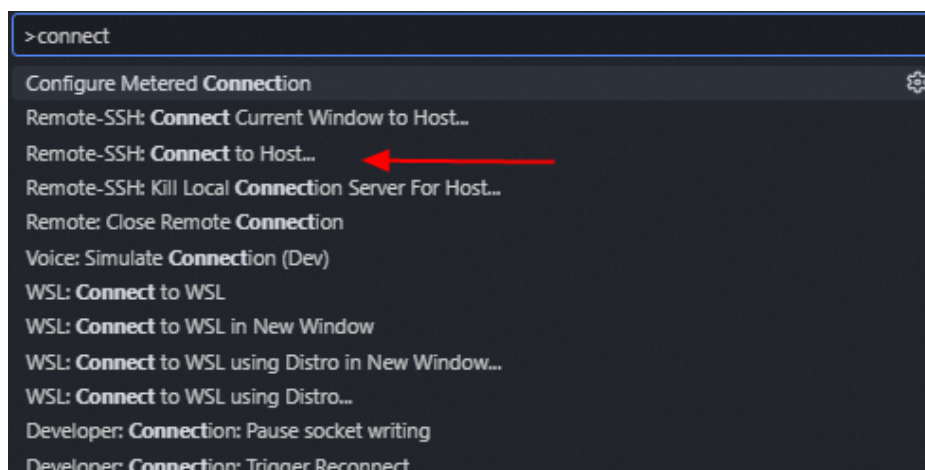
### Tip

The `ServerAliveInterval 15` and `ServerAliveCountMax 4` settings send a keepalive every 15 seconds and disconnect after 4 missed keepalives (60 seconds dead detection). This is critical for proxied connections, which may drop idle sessions after 5–30 minutes.

## 4.5 Connect to the Flexus X via VS Code

### Step by step:

1. Press F1 to open the Command Palette, type **Remote-SSH: Connect to Host...**, and press Enter.



**Figure 5:** VS Code Command Palette — select Remote-SSH: Connect to Host...

2. Select your host name (e.g., `flexusx-dev`) from the list of configured hosts.
3. If prompted to select the platform type, choose **Linux**. This tells VS Code which server binary to download and avoids auto-detection delays.

**Info**

VS Code only asks for the platform type the first time you connect to a new host. The selection is saved in your local settings (`remote.SSH.remotePlatform`) and reused on subsequent connections.

1. VS Code opens a new window and begins connecting. On the first connection, it downloads and installs the VS Code Server on the remote instance — this takes about 1–2 minutes. Subsequent connections are faster.
2. Once connected, click **Open Folder** and select `/home` (or your working directory).
3. Verify the connection by opening a terminal in VS Code (`Ctrl+``) and running:

```
uname -a
df -h
```

**Info**

The first connection installs the VS Code Server on the remote instance, which requires outbound internet access on port 443 (HTTPS). The Flexus X instance has an EIP, so it typically has outbound internet access. If the connection hangs during server installation, verify with `curl -sI pass:https://update.code.visualstudio.com` from the instance.

## 4.6 Troubleshooting

If the connection fails, check these common issues:

- **Ncat: Error reading proxy response** — SSH was not restarted after the port change (Chapter 2). **Fix:** reboot the instance from the Console.
- **Ncat: Error reading proxy response** — proxy `CONNECT` permission missing for this IP. **Fix:** request IT permission for the instance EIP on port 4444.
- **Connection timed out** — proxy unreachable or `CONNECT` blocked. **Fix:** verify proxy address; test `ncat` with the `-v` flag.
- **UNPROTECTED PRIVATE KEY FILE** — key permissions too open. **Fix:** run `icaccls` to restrict the key file to your user only.
- **VS Code hangs on first connect** — instance cannot reach internet on port 443. **Fix:** verify `curl -sI pass:https://update.code.visualstudio.com` from the instance.
- **\$PLATFORM is undefined** — SSH connection failed before server install. **Fix:** check the Remote-SSH output log; fix the underlying SSH error first.

**Info**

To debug the proxy connection, run `ncat` directly with verbose output:

```
"C:\Program Files (x86)\Nmap\ncat.exe" -v --proxy proxy.huawei.com:8080
--proxy-type http <EIP> 4444
```



This shows the CONNECT request and the proxy's raw response (or connection close). For SSH-level debugging, run `ssh -vvv <host-name> "echo ok"`.

| Step       | What to check                                   | Expected result     |
|------------|-------------------------------------------------|---------------------|
| VS Code    | <code>code --version</code> works               | Installed           |
| Remote-SSH | Extension installed                             | Active              |
| SSH config | Host entry with <code>ncat Proxy-Command</code> | Config saved        |
| Connection | Remote-SSH connects to host                     | Flexus X accessible |
| Terminal   | <code>uname -a</code> and <code>df -h</code>    | Output verified     |

**Table 5:** Chapter 4 verification checklist.

# 5

## Install the MaaS Coding Gateway

---

**General Objective:** Install the `oh-my-coding-maas-gateway` on the Flexus X instance. This is a self-hosted LiteLLM proxy that routes Huawei ModelArts MaaS models to coding tools with load balancing, virtual keys, dual-format API endpoints, and Prometheus + Grafana observability.

**Prerequisites:**

- Chapter 4 completed — VS Code connected to the Flexus X via Remote-SSH.
  - Docker and Docker Compose available (the installer handles this).
  - A Huawei ModelArts MaaS API key.
- 

### 5.1 Overview

The `oh-my-coding-maas-gateway` is a self-hosted proxy gateway that sits between your coding tools and Huawei ModelArts MaaS (Model-as-a-Service) models. It provides:

- **Load balancing** — distributes requests across multiple MaaS API keys (N keys → N deployments per model per format).
- **Virtual key management** — each coding tool gets its own virtual key with unlimited budget, isolated from the real MaaS keys.
- **Dual-format API endpoints** — OpenAI Chat Completions (`/v1/chat/completions`) and Anthropic Messages (`/v1/messages`), so tools using different API formats can all reach the same MaaS models.
- **Budget tracking** — per-key spend tracking via LiteLLM PostgreSQL, with cost-per-token configured for all models.
- **Full observability** — Prometheus metrics + Grafana 39-panel dashboard (latency, errors, throughput, tokens, cache, cost).

#### Info

The gateway repository is at [wallacelw/oh-my-coding-maas-gateway](https://github.com/wallacelw/oh-my-coding-maas-gateway). It supports 4 models: `glm-5.2` (1M context), `glm-5.1`, `deepseek-v4-pro`, and `deepseek-v4-flash` (384K max output). This guide was tested with v1.10.9; the installation process is the same for newer versions.

## 5.2 Architecture

The gateway deploys 4 Docker containers via Docker Compose:

| Tools                                                                       | LiteLLM (:4000)          | Huawei MaaS                          |
|-----------------------------------------------------------------------------|--------------------------|--------------------------------------|
| opencode endpoint                                                           | --> /v1/chat/completions | --> openai/ --> MaaS OpenAI          |
| Codex CLI                                                                   | --> /v1/responses        | --> openai/ --> MaaS OpenAI endpoint |
| Claude Code endpoint                                                        | --> /v1/messages/        | --> anthropic/ --> MaaS Anthropic    |
| Pi agent endpoint                                                           | --> /v1/chat/completions | --> openai/ --> MaaS OpenAI          |
| LiteLLM: load-balances across N MaaS keys . PostgreSQL (:5432)              |                          |                                      |
| Observability: LiteLLM --/metrics--> Prometheus (:9090) --> Grafana (:3000) |                          |                                      |

| Service       | Port | Purpose                                 |
|---------------|------|-----------------------------------------|
| LiteLLM Proxy | 4000 | API gateway (OpenAI + Anthropic format) |
| PostgreSQL    | 5432 | LiteLLM database (keys, spend, config)  |
| Prometheus    | 9090 | Metrics storage and querying            |
| Grafana       | 3000 | Dashboard visualization (39 panels)     |

**Table 6:** Docker services and ports.

### Info

All ports bind to 127.0.0.1 (localhost only) by default. To access Grafana and the LiteLLM Admin UI from your local machine, use SSH port forwarding in VS Code or add port forwards in the Remote-SSH connection.

Each model has two deployments — one OpenAI, one Anthropic — so all tools can use the same MaaS models regardless of their API format. The `claude-` prefix (e.g., `claude-glm-5.2`) prevents routing conflicts in LiteLLM.

## 5.3 Run the bootstrap installer

**Objective:** Install the gateway using the one-command bootstrap script.

### Step by step:

1. In the VS Code remote terminal, run the bootstrap script:

```
curl -fsSL https://raw.githubusercontent.com/wallacelw/oh-my-coding-maas-gateway/main/scripts/bootstrap.sh | bash
```

## 2. The installer will:

- Prompt for your Huawei MaaS API key.
- Show an interactive menu to select which coding tools to install (opencode, Codex CLI, Claude Code, Pi agent).
- Auto-install prerequisites (Docker, Node.js, etc.).
- Deploy the proxy + observability stack (LiteLLM, PostgreSQL, Prometheus, Grafana).
- Configure each tool with its own virtual key.
- Run 120 validation checks and offer to install the companion skill.

### Important

Estimated time: ~5 min for a fresh install, ~2 min for an upgrade. If you are behind a proxy, ensure `http_proxy` and `https_proxy` environment variables are set before running the script.

### Tip

For non-interactive installs (CI/automation), pass the API key directly. The `-y` flag suppresses all prompts (prerequisites, tool selection, skill install):

+

```
curl -fsSL https://raw.githubusercontent.com/wallacelw/oh-my-coding-maas-gateway/main/scripts/bootstrap.sh | bash -s -- -y --api-key=sk-xxxx
```

## 5.4 Verify the installation

### Step by step:

1. Check that the proxy is running:

```
curl http://localhost:4000/health
```

2. Open the Grafana dashboard at `http://localhost:3000` to monitor usage and metrics (39 panels across 7 sections).
3. Open the LiteLLM Admin UI at `http://localhost:4000/ui` to manage virtual keys and model routing.
4. Launch your coding tool:

```
opencode # or: codex or: claude --bare or: pi
```



### Tip

To upgrade the gateway later, re-run the same bootstrap command. It detects existing installs, compares versions, and pulls updates — all secrets and data are preserved.

| Step         | What to check                            | Expected result       |
|--------------|------------------------------------------|-----------------------|
| Bootstrap    | Installer completed                      | All services deployed |
| Proxy health | <code>curl local-host:4000/health</code> | Healthy               |
| Grafana      | <code>localhost:3000</code>              | Dashboard accessible  |
| LiteLLM UI   | <code>localhost:4000/ui</code>           | Admin UI accessible   |
| Coding tool  | Launched and connected                   | Gateway routing       |

**Table 7:** Chapter 5 verification checklist.

# 6 Install the Huawei Document Templates Project

---

**General Objective:** Clone the Huawei document templates repository on the Flexus X instance, run the setup script, and verify that the LaTeX toolchain and the coding agent skill are ready for use. The script assumes a fresh Ubuntu install — it installs all dependencies from scratch and configures VS Code at the user level.

## Prerequisites:

- Chapter 5 completed — MaaS gateway installed and running.
- Git installed (`sudo apt install git`).
- The repository URL for the Huawei document templates project.
- A fresh Ubuntu 24.04 instance (no prior LaTeX installation required).

### Info

The `install.sh` script is idempotent — it is safe to run multiple times. It merges VS Code settings rather than overwriting them, and `apt-get install` skips packages that are already present.

## 6.1 Clone and set up the project

### Step by step:

1. Run the one-liner install (clones the repo to `/home/huawei-doc-templates` and installs everything):

```
cd /home
curl -fsSL https://raw.githubusercontent.com/wallacelw/huawei-doc-templates/main/scripts/install.sh | bash
```

### Info

The script installs XeLaTeX, latexmk, fonts (HarmonyOS Sans + Cascadia Code), the open-code skills, and configures VS Code LaTeX Workshop. When run interactively (`./scripts/install.sh`), it prompts for optional components (skills and VS Code — both default to yes). When run via one-liner, all components are installed automatically. It is idempotent — safe to re-run.



**Info**

If an existing installation is detected, the one-liner prompts to update: Update v3.3.6 → v3.3.7? [Y/n]. It pulls the latest version and re-runs the installer. No need to `git pull` manually.

2. Verify the LaTeX toolchain:

```
xelatex --version
latexmk --version
```

## 6.2 Compile the sample guides

The project includes a self-documenting Makefile. To compile all samples (guide + technical + textbook + poc, Portuguese + English):

**Step by step:**

1. From the repository root, run:

```
make samples
```

2. Verify the PDFs were generated:

```
ls -l documents/guide-pt/main.pdf
ls -l documents/guide-en/main.pdf
ls -l documents/technical-pt/main.pdf
ls -l documents/technical-en/main.pdf
ls -l documents/testbook-pt/main.pdf
ls -l documents/testbook-en/main.pdf
ls -l documents/poc-pt/main.pdf
ls -l documents/poc-en/main.pdf
```

**Tip**

To compile a single project, use `make project DIR=documents/guide-pt` or run `latexmk main.tex` from the project's `src/` directory. Run `make` with no arguments to see all available targets.

## 6.3 Create your first guide with the skill

**Step by step:**

1. In the VS Code remote terminal, launch your coding agent:

```
cd /home/huawei-doc-templates
opencode # or: codex, claude --bare, pi
```

2. Run the guide skill:

```
/skill huawei-template-guide
```

3. The skill will ask for:

- **Title** — e.g. “Provisioning an OBS Bucket”
- **Language** — Portuguese or English
- **Project name** — used as the folder name
- **Location** — where to create the project folder

4. The skill generates the .adoc file, creates a .latexmkrc with the correct TEXINPUTS, compiles the document, and reports the page count. **Ready**

### Info

The skill is auto-discovered via `opencode.json` which registers `templates/` as a skill discovery path. No manual skill installation is needed when working inside the repo.

## 6.4 Write content using template commands

The template provides commands for common guide elements:

Headings (automatic numbering, H1 starts on a new page):

```
\section{Chapter Title}          % H1: giant number + title + rule
\subsection{Section Title}      % H2
\subsubsection{Subsection Title} % H3
```

Objectives block (closes with a horizontal rule):

```
\begin{objectives}
  \generalobjective{...}
  \prerequisites
  \begin{itemize}
    \item ...
  \end{itemize}
\end{objectives}
```

Step-by-step and objectives (used inside subsections):

```
\objective{...}
\stepbystep
\begin{enumerate}
  \item ...
\end{enumerate}
```

Callout boxes (environment syntax):

- `warning` — amber warning box
- `tip` — green tip box
- `infobox` — blue info box

Code blocks and inline commands:

- `\begin{code}` — code block (verbatim, no escaping needed inside)
- `\begin{code}[bash]` — same, with a language hint
- `\inlinecode{...}` — inline monospace code

Images, links, and other inline commands:

```
\image{assets/screenshot.png}           % centered image
\imagecap{assets/screenshot.png}{caption} % image with numbered caption
\note{Italic observation.}           % note paragraph
\weblink{https://...}{link text}         % blue clickable link
\menu{VPC, Security Groups}              % menu path: VPC -> Security Groups
\badge{New}                              % inline red label
```

### Info

The project also includes a technical template for technical reports. Use `/skill huawei-template-technical` to create technical reports with a 5-section structure (problem, root cause analysis, root cause, trigger condition, workaround). See `templates/technical/SKILL.md` for the full command reference.

| Step                    | What to check                                     | Expected result            |
|-------------------------|---------------------------------------------------|----------------------------|
| Repository              | Cloned to <code>/home/huawei-doc-templates</code> | Files present              |
| <code>install.sh</code> | Completed without errors                          | Dependencies installed     |
| XeLaTeX                 | <code>xelatex --version</code> works              | Toolchain ready            |
| latexmk                 | <code>latexmk --version</code> works              | Build system ready         |
| Sample PDFs             | <code>make samples</code> compiles all            | Guide + technical, PT + EN |
| Skill                   | <code>/skill huawei-template-guide</code>         | Available                  |

**Table 8:** Chapter 6 verification checklist.

# 7

## Operations and Maintenance

---

**General Objective:** Update, uninstall, or clean up the software and cloud resources created in this guide. All operations in this chapter are optional.

---

### 7.1 Updating the MaaS Gateway

To update the oh-my-coding-maas-gateway to a newer version:

**Step by step:**

1. Re-run the bootstrap script in the VS Code remote terminal:

```
curl -fsSL https://raw.githubusercontent.com/wallacelw/oh-my-coding-maas-gateway/main/scripts/bootstrap.sh | bash
```

2. The installer detects the existing installation, pulls updates, and preserves all secrets, virtual keys, and data.
3. To update only the coding tools (without touching the proxy stack):

```
cd /home/oh-my-coding-maas-gateway  
./scripts/update.sh
```

### 7.2 Updating the Document Templates

To update the Huawei document templates project:

**Step by step:**

1. Re-run the one-liner — it detects the existing installation and prompts to update:

```
curl -fsSL https://raw.githubusercontent.com/wallacelw/huawei-doc-templates/main/scripts/install.sh | bash
```

2. The script pulls the latest version, shows the version change (v3.3.6 → v3.3.7), and re-runs the installer. New templates, commands, or modules are installed automatically.

## 7.3 Uninstalling the MaaS Gateway

To remove the oh-my-coding-maas-gateway:

### Step by step:

1. Run the uninstall script:

```
cd /home/oh-my-coding-maas-gateway
./scripts/uninstall.sh --all --yes
```

2. This stops and removes Docker containers, volumes, and images, removes coding tool configs (opencode, Codex, Claude, Pi), and deletes the repository.
3. To remove only specific components:

```
./scripts/uninstall.sh --tool=opencode    # remove one tool
./scripts/uninstall.sh --docker           # remove Docker stack only
```

## 7.4 Uninstalling the Document Templates

To remove the Huawei document templates installation:

### Step by step:

1. Run the uninstall script (interactive menu with 5 options):

```
cd /home/huawei-doc-templates
./scripts/uninstall.sh
```

2. Or use the one-liner:

```
curl -fsSL https://raw.githubusercontent.com/wallacelw/huawei-doc-templates/main/scripts/uninstall.sh | bash
```

3. The interactive menu offers:

- Option 1: All installed components (skills, modules, font, VS Code)
- Option 2: Everything + apt packages (WARNING: breaks other TeX)
- Option 3: Delete repository directory (all files, guides, documents)
- Option 4: Remove 100% — everything + apt + repo (nuclear option)
- Option 5: Choose specific components individually

4. To preview what would be removed before running:

```
./scripts/uninstall.sh --all --dry-run
```

5. To also remove apt packages (texlive, latexmk, fonts, pandoc):

```
./scripts/uninstall.sh --all --packages --yes
```

### Important

Removing apt packages will break any other LaTeX documents on the instance. Only use `--` packages if you are decommissioning the instance.

## 7.5 Deleting cloud resources (optional)

If you no longer need the Flexus X instance, delete it to stop incurring charges:

### Step by step:

1. In the Console, navigate to **Flexus X**.
2. Select the instance `flexusx-dev`, click **More** and choose **Delete**.
3. Confirm deletion. The EIP is released automatically (configured in Chapter 1 to release on instance deletion).
4. Navigate to menu:VPC[Security Groups], select `sg-xgate`, and delete it.
5. Remove the SSH config entry for `flexusx-dev` from your Windows SSH config file (`C:\Users\<user>\.`

### Important

If you used Terraform to provision the infrastructure, run `terraform destroy` instead to remove all resources declared in the configuration.

| Step                | What to check                                                              | Expected result          |
|---------------------|----------------------------------------------------------------------------|--------------------------|
| Gateway update      | Re-run bootstrap                                                           | New version running      |
| Templates update    | <code>git pull &amp;&amp;&amp;</code><br><code>./scripts/install.sh</code> | New templates available  |
| Gateway uninstall   | <code>./scripts/uninstall.sh --all</code>                                  | Docker stack removed     |
| Templates uninstall | <code>./scripts/uninstall.sh --all</code>                                  | Skills + modules removed |
| Cloud cleanup       | <code>flexusx-dev</code> deleted                                           | No running instance      |
| EIP                 | Released                                                                   | No public IP charges     |
| Security group      | <code>sg-xgate</code> deleted                                              | No SG charges            |
| SSH config          | Host entry removed                                                         | Clean config             |

**Table 9:** Chapter 7 verification checklist.

**Note:** This guide was generated using the `guide.cls` template. For the full command reference, see the template's `README.md` or run `/skill huawei-template-guide` in your coding agent.

# 8

## Brand Color Palette

The template defines auxiliary brand colors aligned to the Huawei Cloud Brand Guidelines (Section 2.12):

| Color    | HEX     | Pantone | Sample                                                                              |
|----------|---------|---------|-------------------------------------------------------------------------------------|
| Rose Red | #C40054 | 7636    |  |
| Dark Red | #7F0001 | 483C    |  |
| Orange   | #ED6D00 | 165C    |  |
| Yellow   | #FCC800 | 7406C   |  |
| Green    | #62B230 | 3501C   |  |
| Blue     | #30B5C5 | 2227C   |  |

The monochrome palette provides a grayscale scale for code blocks and secondary text:

| Color    | HEX     | Brightness |
|----------|---------|------------|
| Black    | #000000 | 0%         |
| Gray 90% | #E5E5E5 | 90%        |
| Gray 80% | #CCCCCC | 80%        |
| Gray 50% | #808080 | 50%        |
| Gray 30% | #4D4D4D | 30%        |
| White    | #FFFFFF | 100%       |

**6.19.2***2026-10-02*

- POC signature cells are more compact: the manual signing blank shrinks from 1.6cm to 1.0cm and cell padding is tighter, so a full 5-row signature grid shares one page with its section heading. The e-mail line is now conditional: label and address share a line when they fit the cell, otherwise the address moves to its own full-width line (keeping the mid-token-break fix).

**6.19.1***2026-10-02*

- POC signature cells: the e-mail address now sits on its own line below the E-mail label (both `\signaturecell` and `\signedcell`), giving long addresses the full column width and fixing mid-token breaks; DOCX/MD/HTML parity.

**6.19.0***2026-10-02*

- POC template: new `\signedcell` macro for pre-signed signature cells (sign-off already recorded, for example an electronic signature – no manual signature blank), alongside the existing `\signaturecell`. Language-aware label (Signed by / Assinado por); DOCX/MD/HTML parity; demonstrated in both POC samples.

**6.18.0***2026-09-29*

- Header title now defaults to the document title when `:header-title:` is unset (PDF); the DOCX header honors `:header-title:` (literal text replaces the `STYLEREf` field), so both formats behave identically.

**6.17.0***2026-09-29*

- Multi-logo support: `:extra-logo-1:` and `:extra-logo-2:` render in the header right corner and on the cover as an equal-height row; `:cover-logo-height:` overrides the row height; DOCX parity including document-relative asset paths.

**6.16.4***2026-09-28*

- Testbook: added colored badge column to acceptance criteria table, matching badges used in test-cases and test summary. Updated SKILL.md skeleton with badge example.

**6.16.3***2026-09-28*

- Oracle review fixes: corrected SKILL.md code example (`fancyvrb` multi-line), fixed typo in testbook-pt, synced `stripebg` to DOCX/HTML, added `stripebg` to all SKILL.md color tables, fixed stale field order and comments.



## 6.16.2

2026-09-28

- SKILL.md (testbook): documented auxiliary LaTeX commands available in passthrough blocks (`\menu`, `\inlinecode`, `\param`, `\weblink`, `\note`, `\image`, `\imageplaceholder`, `\code`, `\warning`). Added `stripebg` color (`#F0F1F3`) for visible table zebra striping. Documented `noanswers` option usage. Added sample-only notes to testbook demo sections.

## 6.16.1

2026-09-28

- Fix: H1 section heading red rule now spans full page width (reverted minipage wrap that constrained `\titlerule`). Cover text casing “Co., Ltd.” synced to all samples. Testbook: removed duplicate prerequisites from objectives block.

## 6.16.0

2026-09-26

- Visual polish and functional improvements: cover composition (`\vfill` anchors identity block to lower third); widow/orphan control; H1 long-title wrap safety (minipage); code block continuation indent; roman-numeral list alignment; cover text casing (Co., Ltd.); teststep hanging indent; testsummary longtable breakability + zebra striping; POC signature signing line; checkbox size; stakeholders name hyphenation fix.

## 6.15.1

2026-09-26

- TOC fix: two-digit section numbers (10+) no longer collide with titles (widened `\cftsecnumwidth`); front matter now uses roman page numbering (i, ii) so the TOC no longer shows “Page 2” followed by body “Page 1”.

## 6.15.0

2026-09-26

- Visual verification standard: `scripts/render-pages.sh` renders PDF/DOCX/HTML to PNG page images; AGENTS.md documents the render `-> @observer ->` reconcile workflow with a defect checklist, cross-format rule, and known DOCX tolerances.

## 6.14.1

2026-09-26

- Badge pills no longer overflow their frames: `\huaweibadge` now auto-expands for labels wider than the nominal width (matching the DOCX PNG min-width behavior), and the testbook testsummary result column widens to the measured pill widths (4.5cm PT, 2.7cm EN), eliminating the badge-row Overfull warnings.

## 6.14.0

2026-09-23

- Converter content-loss fixes: compound admonitions no longer corrupt nested tables and code, definition lists keep their definitions, colspan is supported (rowspan warns), block anchors on tables/images fix dead cross-references, and technical cover values (the `setreport` commands) now render on the PDF cover. Empty-macro tests fixed with `etoolbox`’s `ifdefempty` (covermeta shows the version; phantom Author row gone). HTML: double captions eliminated, EN admonition titles match the PDF (Info/Important/Tip), and technical sections render as language-aware headings in DOCX/MD/HTML. Build: font-fallback detection fixed (was dead code); `install.sh` now installs `asciidoctor-diagram` so fresh installs no longer lose diagrams silently. Round-trip heading checks are now exact.

### 6.13.0

2026-09-23

- Test-result vocabulary unified: Portuguese badges now read *Atende / Atende com ressalvas / Não atende / Não testado* in both testbook and POC (was *Aprovado/Reprovado/Bloqueado/Não testado* and *Atendido/Parcial/Falha/Ignorado*). Testbook gains the Partial result (*Atende com ressalvas*, orange pill); Blocked was dropped (legacy documents render the neutral fallback pill). English display and source enum values unchanged. Existing PT documents render the new names on recompile.

### 6.12.0

2026-09-23

- Language audit: Portuguese documents no longer show English labels. Technical cover labels (*Versão/Data/Cenário/Autor*), testbook badges (*Aprovado/Reprovado/Bloqueado/Não testado*), HTML admonition titles and captions, DOCX callout labels, General Objective, E-mail, and image placeholder are now language-aware under `:lang: pt`. CSS technical section headings gained PT selectors; 6 CSS strings aligned with PDF. Badge PNGs auto-sized to fit text. Sample wording fixed: *Cor/Amostra* table headers, *Infraestrutura como Código*, *provedor* heading.

### 6.11.1

2026-09-22

- Documentation: document lifecycle clarified. Documents created via the skills are standalone deliverables — git is never used for individual documents; the `:version:` attribute and changelog block are the document's own version control. All four SKILL.md files gained a "Document lifecycle" section and now scope AGENTS.md's git workflow to template repository development only. AGENTS.md gained a Scope section stating user documents are gitignored and never repository content, and L17 is clarified to apply to template releases only.

### 6.11.0

2026-09-22

- Project-wide audit: converter robustness and cross-format consistency. Objectives roles now work in all forms — inline (`[.general-objective]` and `[.objective]`) and block (inside `[.objectives]`) — rendering language-aware labels (the block form previously crashed the build; the inline form silently dropped the label). The `[.prerequisites]` and `[.stepbystep]` roles attached to lists no longer drop items. Multi-author documents now list all authors. Conversion errors print a clean file:line message instead of a Ruby object dump. Unknown code languages (`go`, `java`, `rust`, `terraform`, ...) fall back to plain verbatim with a compile warning (was a hard error). DOCX/MD/HTML: codefile blocks (`[.codefile,file=...]`) now inline the file content like the PDF, and admonition labels no longer corrupt NOTE:/TIP: lines inside code blocks. DOCX Fail/Falha badge backgrounds and plain-table borders now match the PDF colors. Shared class boilerplate extracted into the new `huawei-base.cls` base class (rendering unchanged). Watch mode survives compile errors; install reports failures instead of silent success; the `/etc/LatexMk` write is opt-in; build diagnostics are kept on failure; asciidoctor warnings now fail the build; a missing header logo warns instead of erroring. New `test-pdf-compile.sh` replaces the legacy `test-filter` stub; `round-trip` now validates the production build pipeline. Documentation corrected across README, AGENTS, and all SKILL files (objectives syntax, codefile example, supported languages, longhutable restriction, install requirements, pandoc range, security note).

### 6.10.0

2026-09-21

- Header logo, callout boxes, and note labels: DOCX header now shows the Huawei logo in the top-left corner (optional, driven by the `:header-logo:` attribute). DOCX admonitions (TIP, NOTE, WARNING) now render as styled callout boxes with colored backgrounds, left borders, and type labels —

matching the PDF. PDF inline notes ([.note] role) now show a language-aware “Note:” label (was unlabeled italic text).

## 6.9.2

2026-09-21

- Fix DOCX body text alignment: paragraphs are now justified (matching the PDF LaTeX default), instead of left-aligned.

## 6.9.1

2026-09-21

- Fix DOCX subsection number size: H2/H3/H4 section numbers were rendered at 56pt (same as H1) instead of matching the heading text size (18/16/14pt). The SectionNumber style is now unstyled (inherits heading size); 56pt is set directly on H1 number runs only.

## 6.9.0

2026-09-21

- TOC styling: the table-of-contents title rule is now Huawei red (was black) in both PDF and DOCX, matching the H1 heading red rules. DOCX TOC entries now match the PDF: level 1 bold 12pt, levels 2-3 normal 10pt, HarmonyOS Sans font. DOCX TOC depth increased from 2 to 3 levels (matching PDF tocdepth). Dead TOCTitle style removed from DOCX.

## 6.8.3

2026-09-20

- Docs accuracy fix: the multi-format pipeline descriptions in README, AGENTS, and all four SKILL.md files now correctly separate the DOCX/Markdown path (pre-processor then docbook then pandoc) from the HTML path (pre-processor then asciidoctor html5) — the previous wording implied HTML also went through docbook and pandoc. Corrected a stale comment in the DOCX fixer.

## 6.8.2

2026-09-20

- Documentation and cleanup: README, AGENTS, and all four SKILL.md files now describe the real multi-format pipeline (pre-processor + docx\_fix -fix + -lang flag) instead of the pre-v6.4 asciidoctor-to-pandoc chain. DOCX TOC title is now right-aligned like the PDF (was targeting a dead style). Signature names render bold like the PDF. Dead code removed (unused Caption style, dead TOCTitle style, dead HTML badge CSS). The -template and -lang flags accept the equals form (-template=guide). Various stale comments corrected.

## 6.8.1

2026-09-20

- Quality-pass fixes for v6.8.0: non-technical covers no longer show a stray Date paragraph (regression from the meta-scan change); the technical cover table is no longer hijacked from the body when :nochangelog: is set (boundary check on the cover-table detection); the Version/Date/Scenario table now renders unconditionally like the PDF (was gated by :nochangelog:); empty cover tables are skipped when no report values are set; corrected the cover-element-order wording in the v6.8.0 changelog.

## 6.8.0

2026-09-20

- Technical report covers now match the PDF in DOCX/MD/HTML: the title uses the PDF’s 24pt size (was 36pt), the report type label appears below the title (the DOCX cover keeps the established title-first order; the PDF has label-then-title), and the Version/Date/Scenario table is rendered with

the PDF's red label column. The cover meta line now shows the report version and its set date (was the document version and the build date). Authors, when set, appear as a table row like the PDF.

### 6.7.0

2026-09-20

- Badge and language fidelity: DOCX badge pills now render black text like the PDF (was frame-colored), use the correct label case (Pass, not PASS), include the Portuguese POC labels (Atendido, Parcial, Falha, Ignorado), and render at native width per template (testbook badge text no longer shrinks). Portuguese signature greetings now match the PDF (At.te.), the POC classification label matches the full PDF wording, and the secondary-format table headers (stakeholders, closing record, test summary, changelog) are now language-aware. DOCX footers show Página in Portuguese documents. Cover dates no longer zero-pad the day.

### 6.6.1

2026-09-20

- Quality-pass fixes for v6.6.0: the TESTCASE-marker strip now covers the entire DOCX fix run (a failure at any stage can no longer leave markers in the output file); the fixer resets its template state between in-process runs (the documented path-sniffing fallback works again); build.sh also cleans temporary files on interrupt signals; fixed a CHANGELOG typo.

### 6.6.0

2026-09-20

- Pipeline robustness: DOCX post-processing now fails loudly when content styling errors occur (TESTCASE markers can no longer leak into output); the DOCX fixer takes an explicit `--template` flag so badge sizing no longer sniffs the output path; build.sh cleans temporary files on all exit paths and makes the asciidoctor-diagram dependency optional for HTML generation; the pandoc version pin now warns instead of failing hard.

### 6.5.2

2026-09-20

- Quality-pass fixes for v6.5.1: LaTeX quote conversion now handles only paired quotes (quote characters inside code arguments are no longer corrupted); signature cells containing escaped ampersands are no longer split; round-trip HTML generation guards the asciidoctor-diagram dependency and fails loudly when generation fails; pre-processor unit tests now cover pipe escaping in all five table handlers.

### 6.5.1

2026-09-20

- Fixed DOCX/MD/HTML output: tables are no longer corrupted when changelog, signature, or summary content contains a pipe character (“|”). Badge text is now preserved in DOCX/MD/HTML — a “Ready” badge renders “Ready” instead of “[NEW]”. Multi-item changelog entries render as bullet lists instead of being joined with a literal “+”. POC documents now include the changelog in DOCX/MD/HTML. The :noanswers: attribute now hides test results and remarks in DOCX/MD/HTML, matching the PDF. Inline LaTeX (monospace text, arrows, LaTeX quotes) no longer leaks from changelog entries. The test suite now exercises the real build pipeline (pre-processor wired into round-trip and DOCX tests), with new pre-processor unit tests.

### 6.5.0

2026-09-20

- Evidence checklist simplified to plain bullets in both PDF and DOCX (checkbox markers removed for compatibility). Planned Activities list in DOCX now uses Word's native lower-roman numbering

(i., ii., iii.) matching the PDF. Badges in DOCX replaced with exact-replica PNG pill images (rounded, colored frame and background, bold text) matching the PDF \huaweibadge look — assets committed to the repo. Signature blocks in DOCX no longer show literal + symbols (proper hard line breaks).

#### **6.4.7**

2026-09-20

- Section titles in DOCX now match the PDF heading design: the big section number (56pt) sits on the left, the title text is pushed to the right edge (a right tab stop mimics the LaTeX \hfill), and the red rule runs below — previously the title sat left-aligned next to a small number. Applies to all templates.

#### **6.4.6**

2026-09-20

- Code blocks inside test cases no longer break the red left rule in DOCX — the code style's left indent (which jogged the rule) is reset inside test case blocks, and code blocks now show the PDF's code box (gray background with thin border on top, bottom, and right, red rule continuing on the left).

#### **6.4.5**

2026-09-20

- DOCX polish pass: all captions centered (matching the PDF), vertical spacing between blocks matched to the LaTeX rhythm (medskip/smallskip around captions, field bars, and steps), and test steps now render as separate paragraphs each with the red bold number (previously merged into one paragraph).

#### **6.4.4**

2026-09-20

- DOCX covers now show the logo, cover text, and version/date/time meta line like the PDF. Test case blocks in DOCX get the PDF's visual wrapper: red left rule, red field header bars, and red step numbers. Captions follow the PDF ordering system in DOCX, MD, and HTML (Table/Diagram/Figure/Test-case N: with bold symbol). Portuguese PDFs now label diagrams "Diagrama N:" (was hardcoded English "Diagram"). Markdown and HTML output no longer leaks raw LaTeX from passthrough blocks — the DOCX pre-processor now feeds those pipelines too.

#### **6.4.3**

2026-09-20

- Testbook and POC DOCX output now matches the PDF much more closely: native table of contents (language-aware, auto-populated), each section starts on a new page, embedded images and diagrams (PlantUML/Graphviz rendered), white-on-red table headers, cover version line under the title, centered test case captions, and plain black rules for signature grids.

#### **6.4.2**

2026-09-20

- DOCX output for POC and testbook templates now includes all passthrough content (stakeholders, closing record, signatures, test cases, test summary, changelog) via a new pre-processor. Fixed dropped Test Scope fields in testbook DOCX output.

#### **6.4.0**

2026-09-20

- Fixed changelog dates — replaced \today with historical release dates. Eliminated zero-width-space missing-glyph warnings from typographic substitutions.

### 5.1.1 2026-09-14

- Synced callout colors in DOCX/HTML/MD output to match brand palette. Added color palette demo.

### 5.1.0 2026-09-14

- Adopted brand guidelines auxiliary color palette and monochrome scale. Aligned callout colors to brand specifications.

### 5.0.3 2026-09-14

- Removed non-standard `\badge` tags from testcases for consistency.

### 5.0.2 2026-09-14

- Long commands in test steps now render as code blocks instead of inline code.

### 5.0.1 2026-09-14

- Testcase caption centered above the block.

### 5.0.0 2026-09-14

- Major refactoring: extracted shared modules (huawei-titles, huawei-toc, huawei-cover), renamed `\lg@guidetitle` to `\lg@doctitle`, unified title API with `\setdoctitle`, decomposed `docx_fix.py` (680→84 lines), added testbook unit tests, fixed build system error handling.

### 4.7.0 2026-09-14

- Testcase headings now formatted as captions (like figures/tables): bold “Testcase N:” followed by description. Changed label from “Test Case” to “Testcase”.

### 4.6.0 2026-09-13

- Section restructure: Introduction, Test Cases (with category subsections), Conclusion. Testcases auto-numbered like figures/tables. Changed testcase from subsection to subsubsection.

### 4.5.0 2026-09-13

- Unified field APIs: `testprerequisites` and `testexpected` changed from commands to environments, matching `testprocedure`. All three now use `\teststep` for auto-numbered items with red bold numbers. Uniform code, same visual style.

### 4.4.0 2026-09-13

- Code review fixes: removed dead labels, badge now uses Huawei brand colors (`tipbg/tipfg`, `huaweired`, `warningbg/warningfg`), Lua filter preserves formatting in DOCX/MD/HTML via `parse_latex_blocks`, `testprocedure` handler preserves non-step content (images, code, callouts).

### 4.3.0 2026-09-13

- Added `testlist` environment for numbered lists with red bold numbers in testcase fields. Fixed alignment: zeroed `\parindent` inside testcase, indented procedure step numbering to align with `testlist` items.

#### 4.2.0

2026-09-13

- Changed teststeps from tabular table to auto-numbered paragraphs. `\teststep` now takes 1 arg (action only, numbering automatic). Removed teststeps environment. Images, code blocks, and callouts can be placed freely between steps inside testprocedure.

#### 4.1.0

2026-09-13

- Testcase field redesign: full-width red mini header bars for each field. Changed testprocedure from command to environment wrapping 2-column teststeps (Step | Action). Reordered fields: Test Result before Remarks. Updated Lua filter and round-trip test for grid table counting.

#### 4.0.0

2026-09-13

- Redesigned testbook testcase environment: replaced 2-column tabular with breakable tcolorbox (red left-rule, stacked paragraph fields). Images, code blocks, callouts, and nested tables now render correctly and break across pages. Added longhutable environment for page-breaking tables. Updated Lua filter to produce definition lists for DOCX/MD/HTML.

#### 3.9.0

2026-09-13

- Added testsummary environment, teststeps environment, and `\testresultbadge` command to testbook template. Fixed testcase layout (full-width tabular with proper text alignment).

#### 3.8.1

2026-09-13

- End-to-end quality review: fixed testbook PDF metadata title, removed stray user DOCX from template dir, fixed round-trip.sh variable typo, updated stale references in AGENTS.md and README.md, committed testbook sample PDFs, set TZ for pandoc cover time consistency, widened pandoc version range.

#### 3.8.0

2026-09-13

- Removed Tips and Tricks chapter (tmux configuration) — required VS Code-specific keybindings that don't generalize across terminal setups.

#### 3.7.0

2026-09-12

- Added testbook template for POC/acceptance test case documents with testcase environment and `[noanswers]` option.

#### 3.6.0

2026-09-11

- Added `\setdocauthors` command and `[noauthors]` class option to both templates. Authors are displayed on the cover page (optional — hidden if not set or if `[noauthors]` is passed). Demonstrated in all 4 samples.

#### 3.5.0

2026-09-11

- Reorganized project structure: moved `build.sh`, `install.sh`, and `uninstall.sh` from repo root to `scripts/` directory. Updated all references in Makefile, README, AGENTS.md, SKILL.md, and setup guide. One-liner URLs now use `main/scripts/install.sh`.

### 3.4.0

2026-09-11

- `install.sh`: `opencode skills` and `VS Code LaTeX Workshop` are now optional prompts (default yes). Removed `LTεX` extension (was failing to install). One-liner detects existing installation and prompts to update with version display (`v3.3.6` → `v3.3.7`). Uses absolute paths to prevent nested clones.
- `uninstall.sh`: Added one-liner support (`curl | bash`). Added “Remove 100%” nuclear option (option 4) to interactive menu. Added `--repo` flag.
- Updated Chapter 6 with optional component prompts and update detection. Updated Chapter 7 with one-liner update/uninstall and full menu documentation.

### 3.3.0

2026-09-11

- Moved pre-compiled setup guide to `setup-guide/` folder in repo root with all four formats (PDF, MD, DOCX, HTML). Users can read these before cloning to understand the installation process.
- Updated Makefile `setup-guide` target to generate and copy all four formats.
- Updated README with pre-compiled setup guide section and format comparison table.
- Updated `.gitignore` to track all four formats in `setup-guide/`.

### 3.2.1

2026-09-11

- Renamed Chapter 7 from “Clean Up” to “Operations and Maintenance”. Added subsections for updating and uninstalling both the MaaS Gateway and the document templates. Added cloud resource cleanup as an optional subsection.

### 3.2.0

2026-09-11

- Added `uninstall.sh` script with interactive menu, `--all`, `--yes`, and `--dry-run` flags. Removes `opencode skills`, `.sty` modules, `HarmonyOS Sans` font, `/etc/LatexMk` fix, and `VS Code` settings.
- Fixed `install.sh`: show `apt-get` output (was hidden), suppressed `fvextra` build noise, fix font detection (`fc-list : family`), fix test compilation paths (`src/` subdirectory), fix `LTεX` extension success/failure reporting.
- Updated Chapter 7 with uninstall instructions.

### 3.1.2

2026-09-11

- Added output format guide to `README.md`: comparison table of PDF, Markdown, DOCX, and HTML with purpose, copy-paste quality, and limitations. Recommends Markdown for copy-paste.

### 3.1.1

2026-09-11

- Updated MaaS Gateway chapter to v1.10.9: non-interactive mode (`-y`) now suppresses all prompts (prerequisites, tool selection, skill install).
- Changed project clone path from `~` (root home) to `/home` as default working directory.
- Added project standards to `AGENTS.md`: workflow, end-to-end validation, code style, git conventions, when unsure.

### 3.1.0

2026-09-11

- Switched from ECS to Flexus X instance (4 vCPU, 16 GB RAM via slider, flavor `x1.4u.16g`) — more cost-efficient and robust. Updated creation steps to match Flexus X console order: billing mode



first, vCPU/RAM slider (no flavor dropdown), Cloud Eye monitoring (recommended), then instance name and key pair login. Replaced screenshots with Flexus X console images.

- Rewrote Chapter 3: switched from `connect.exe` to `ncat` (from Nmap) as the SSH ProxyCommand. Added explanation of HTTP CONNECT method, why port 4444 works, why `ncat` is chosen, `ncat` installation with antivirus exclusions, flag reference table, and improved keepalive settings.
- Restructured chapters: moved SSH config and VS Code connection to new Chapter 4 (Install VS Code and Connect via Remote-SSH). Chapters 5–7 renumbered.
- Added verification checklist tables to all 7 chapters.
- Added mandatory reboot warning in Chapter 2 (SSH port change). Added troubleshooting subsection in Chapter 4 with common SSH connection issues and `ncat` debugging commands.

### 3.0.1

2026-09-10

- Minor fixes: `test-filter.sh` auto-discovery, `build.sh` help text, Makefile aggregate descriptions, 5-section comment correction, CHANGELOG version gap note, callout-in-code fix, CI font auto-discovery, `dofile` path comment, `test-sync.sh` header.

### 3.0.0

2026-09-09

- Modular pipeline refactoring: shared Lua filter factory, shared DOCX fix logic, build system auto-discovery, unified format pipeline, GitHub Actions CI.

### 2.16.0

2026-09-05

- Updated to match template v2.16.0 — Lua filter fixes, build system updates, and technical template parity improvements.

### 2.13.0

2026-08-24

- Foundation refactoring Phase 4: make DOCX `--fix` post-processing robust. Replaced all silent-skip guards in `create-reference-docx.py` with loud `RuntimeError` assertions for expected XML elements (Heading1–4, Title, VerbatimChar, Normal, docDefaults, numbering indentation). Added pandoc version pin (3.1.0–3.2.0) to catch output-structure regressions on upgrade. Extended `test-docx-fix.sh` with version check and 4 loud-failure tests (missing style → non-zero exit).

### 2.12.0

2026-08-24

- Foundation refactoring Phase 3: introduce `parse_preamble()` in the Lua filter, extracting all class options (`portuguese`, `indentbody`, `notime`, `nochangelog`) and all `\set*` command values into a single preamble state object. Fixes 6 L18 divergences at root cause: English TOC label (“Table of Contents” → “Contents”), `[notime]` now gates time in DOCX/HTML, `[nochangelog]` now suppresses changelog in non-PDF formats, `\setcoverlogo` honored instead of hardcoded, `\setheaderlogo` warns if set in non-PDF, `\note` renders as italic in all formats (matching PDF). Any future preamble command or class option now flows through automatically.

### 2.11.0

2026-08-24

- Foundation refactoring Phase 2: extend test matrix to cover all 3 output formats × 3 documents (9 cells). `round-trip.sh` now generates DOCX, counts structures in `word/document.xml`, and checks cross-format consistency (headings, images, code blocks, tables, callouts, 0 raw LaTeX across MD+DOCX+HTML).

Added `test-docx-fix.sh` smoke test asserting H1 red border, heading colors, list indentation, and footer PAGE field in patched `styles.xml`.

## 2.10.0

2026-08-24

- Foundation refactoring Phase 1: sync cls version to project tag, fix stale doc references (L16 output count, L-range, setup-guide feature claim), add `test-sync.sh` for version consistency checking.

## 2.9.1

2026-08-23

- HTML and Markdown outputs now embed images as base64 data URIs. HTML uses `pandoc --embed-resources`, MD uses post-processing with `embed-images.py`. Outputs are self-contained — no external image files needed.

## 2.9.0

2026-08-23

- DOCX: fix content width (8504→9638tw, margins are 2cm not 3cm). Add footer page numbers. Style TOC heading (22pt bold + rule). Add hutable table width + fixed layout.

## 2.8.3

2026-08-23

- DOCX fix: callout tables now have explicit width (8504tw) and fixed layout. Changelog section heading now uses custom H1 format (56pt number, red border, right tab) instead of pandoc's default.

## 2.8.2

2026-08-23

- DOCX H1 heading: reverted table approach to paragraph with tab stop + red bottom border. Table broke navigation pane and TOC field. Paragraph preserves Heading1 style for navigation while matching PDF layout (number left, title right, red rule below).

## 2.8.1

2026-08-23

- DOCX heading fix: text color reverted to black (was incorrectly changed to Huawei red in v2.8.0). H1 now uses a two-column table (number left, title right) with red bottom rule. H2-H4 use simple inline format (number + title, no right tab stop).

## 2.8.0

2026-08-23

- DOCX output: match PDF styling — heading colors (Huawei red), H1 number size (56pt), callout borders (left-only) and padding, changelog formatting (rules, bold version, italic date), caption style (9pt bold), badge style (8pt bold white on red), list indentation.

## 2.7.1

2026-08-17

- Rename project repository to all lowercase (`huawei-doc-templates`).

## 2.7.0

2026-08-12

- The `changelog` environment now emits its own section heading; `[nochangelog]` suppresses the heading and entries in one switch.

## 2.6.1

2026-08-12

- Fix `hutable` centering and row-color fill (switched to `\hline` for `colortbl` compatibility).

**2.6.0** 2026-08-12

- Switched verification table to the new `hutable` full-grid style.
- Floats now default to `[H]` placement (in-source order).

**2.5.1** 2026-08-10

- H1 section title size reduced from 27pt to 20pt.

**2.5.0** 2026-08-10

- Default image size increased: width 65% → 90% of text width, height 40% → 50% of text height.

**2.4.0** 2026-08-10

- Enlarged H1 section title (18pt → 27pt, 50% bigger).
- `nochangelog` option now hides version, date, and time on the cover page.

**2.3.1** 2026-08-09

- Rolled back `/ActualText` markers — broke code block rendering.

**2.3.0** 2026-08-09

- Added key-value sizing options to `\image` and `\imagecap` — width and height can now be set independently.

**2.2.4** 2026-08-09

- Fixed PDF copy-paste of URLs in code blocks — disabled Cascadia Code programming ligatures (`ca1t`) that corrupted `//` into `))` on copy.

**2.2.3** 2026-08-09

- Fixed italic font rendering — HarmonyOS Sans and Cascadia Code have no italic variants; enabled synthetic slant (`AutoFakeSlant`).

**2.2.2** 2026-08-08

- Fixed chapter reference in prerequisite (Chapter 3 → Chapter 4).

**2.2.1** 2026-08-08

- Changed table and figure caption labels to black (was Huawei red).

**2.2.0** 2026-08-08

- Added Huawei-branded table styling: red header bar, alternating row colors, and bordered cells.

**2.1.0** 2026-08-06



- Updated ECS flavor to `ac8.xlarge.2` (4 vCPU, 8 GB RAM).
- Updated OS image to Ubuntu 24.04 Server 64bit.
- Increased disk size to 100 GB.
- Recommended pay-per-use billing mode.
- EIP auto-assigned: pay by traffic, 1000 Mbit/s, release with ECS.
- Key pair authentication (created in Huawei Cloud); root password can be set later as a second access method.
- Security group `sg-xgate` now created **before** the ECS instance.
- Replaced SSH port 22 with port 4444 (Huawei proxy blocks port 22).
- Restricted SSH access to three specific Huawei internal IPs (119.8.89.193, 119.8.89.3, 119.8.193.3).
- Added Port 4444 to SSH config examples.

## 2.0.0

2026-08-05

- Replaced OpenCode with `oh-my-coding-maas-gateway` as the default coding environment.
- Added changelog and versioning support to the template (`\setdocversion`, `\setdocdate`, `changelog` environment).
- Renamed `info` environment to `infobox` to avoid package name collisions.
- Fixed external file code block implementation (`\codefile`).

## 1.0.0

2026-08-05

- Initial version.
  - Added ECS provisioning on Huawei Cloud.
  - Added SSH with proxy configuration for VS Code Remote-SSH.
  - Replaced OpenCode with `oh-my-coding-maas-gateway` (LiteLLM proxy with load balancing, virtual keys, and Grafana observability).
  - Added Huawei document templates project setup.
  - Added changelog and versioning support to the template.
-